GE VERNOVA

ADMS

16.25.1

AMI Server - Landis and Gyr Command Center Interface Integration Guide

Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 4, 2024	ADMS 16.7.0: • Updated document format and styling.

Contents

1. Introduction

The Landis and Gyr (L+G) interface definition is based on the CIM 2.0 message format. Message transfer is completed using SOAP over HTTP. Security is enabled through a combination of SSL, basic authentication, and authorization headers.

The ADMS AMI Server interface supports the L+G CIM first and second edition interfaces. Each service runs under a separate endpoint; however, the functionality is contained within a single instance of AMI Server.

1.1. Supported Landis and Gyr Functions

The following Landis and Gyr events are supported by the ADMS AMI Server:

- Meter pings
- Power on/off exceptions
- Voltage sag/swell exceptions
- On-demand readings

1.1.1. Meter Pings

Meter pings are initiated by ADMS and sent to L+G Command Center. Each request contains a single meter. L+G Command Center responds immediately with the request status. Once the meter ping completes, L+G Command Center replies with the ping result.

1.1.2. Power On/Off Exceptions

Power on/off exceptions are sent from L+G Command Center as soon as the event occurs. These events do not require a request.

1.1.3. Voltage Sag/Swell Exceptions

Voltage sag/swell exceptions are stored and sent at an irregular interval (about once an hour). The "packet" of events are sent from L+G Command Center without a request. ADMS only processes the latest event, so the AMI Server interface groups the events and forwards only the most recent event to ADMS. In addition, the EventValidTime configuration value is used to filter out any events that originated earlier than the specified time, in minutes. For more details, see AMI Server Configuration File.

1.1.4. On-Demand Readings

On-demand reading requests are initiated by ADMS and sent to L+G Command Center. Each request contains a single meter. L+G Command Center responds immediately with a request status. Once the meter reading completes, L+G Command Center replies with the results. Reading types are defined in the CIM IEC 61968-9 standard.

In the CIM 1 interface, reading types are defined in the Command Center configuration and a request from ADMS does not specify individual reading types. In the CIM 2 interface, ADMS requests the desired reading types with each request.

Voltage values are per phase, while kW and kVar values are total. Watt and Var reading are evenly split among phases that report a voltage. In the case where no kVar value is available, AMI Server calculates the kVar using the configured PowerFactor if CalulateVar is set to true. For more details, see AMI Server Configuration File.

2. Configuration

> **Note**
>
> This chapter does not include instructions for installing and configuring the Landis and Gyr interface. The L+G interface should already be set up when you configure the ADMS AMI Server.

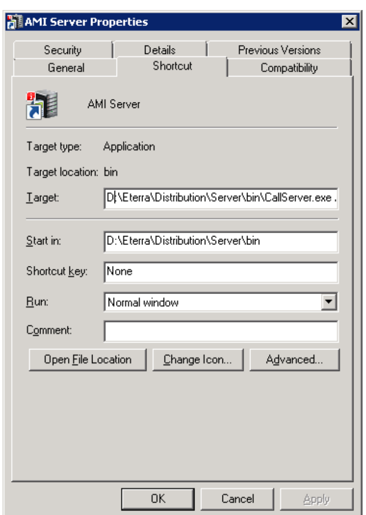
The ADMS AMI Server and components related to the L+G interface are included in the ADMS server installer. The AMI Server is essentially the Call Server, which runs with the `AmiServerConfig.xml` configuration file

2.1. Configuring the AMI Server Windows Shortcut

To configure the AMI Server's shortcut:

1. Navigate to the `\Distribution\Server\bin` folder.
2. Create a new shortcut for the Call Server and set the name to "AMI Server".
3. Set the Target value of the shortcut as follows:

```
D:\Eterra\Distribution\Server\bin\CallServer.exe
/CONFIG%IDMS_ROOT%\config\server\AmiServerConfig.xml
```



You can now start the AMI Server using this shortcut. AMI Server can optionally start in interactive mode, or be configured to run in a PROCMAN setup.

2.2. Configuring Call Server

To configure the Call Server, in the `CallServerConfig.xml` file, make sure the following settings are set to false:

```
<DisableItronInterface>false</DisableItronInterface>
<DemoMeterMode>false</DemoMeterMode>
```

2.3. Configuring AMI Server

The `AmiServerConfig.xml` configuration file is located at `%IDMS_ROOT%\config\server`. To configure the AMI Server, edit the configuration file and customize the `AmiConfig` section, `ApplicationParameters` section, and other parameters as appropriate.

The following sample is an example configuration of key parameters:

```
<ApplicationParameters>
  <EnableAmiInterface>true</EnableAmiInterface>
  <AmiInterfaceName>LGCIM</AmiInterfaceName>
  <AmiInterfaceDllPath>%DMS_BINDIR%LGWebServices.dll</AmiInterfaceDllPath>
  <AmiInterfaceDllTypeName>LGWebServices.LGWebServices</AmiInterfaceDllTypeName>
</ApplicationParameters>
<AmiConfig>
<EnableEventListenerServiceCIM1stEdition>true</EnableEventListenerServiceCIM1stEdition>
<EventListenerServiceAddress1stEdition>http://localhost:8800/CIM1stEdition</EventListenerServiceAddres
s1stEdition>
  <OperationsCIM1stEdition>
    <BaseAddress>http://localhost:5040/ServiceEndpoint</BaseAddress>
    <UserName>user</UserName>
    <Password>password</Password>
    <ReplyAddress>http://localhost:8800/LGCIM</ReplyAddress>
    <UseAppConfig>false</UseAppConfig>
    <CertStoreLocation>LocalMachine</CertStoreLocation>
    <CertStoreName>Root</CertStoreName>
    <CertFindType>FindBySubjectName</CertFindType>
    <CertFindValue />
  </OperationsCIM1stEdition>
<EnableEventListenerServiceCIM2ndEdition>true</EnableEventListenerServiceCIM2ndEdition>
<EventListenerServiceAddress2ndEdition>http://localhost:8800/CIM2ndEdition</EventListenerServiceAddres
s2ndEdition>
  <OperationsCIM2ndEdition>
    <BaseAddress>http://localhost:5040/ServiceEndpoint</BaseAddress>
    <UserName>user</UserName>
    <Password>password</Password>
    <ReplyAddress>http://localhost:8800/LGCIM</ReplyAddress>
    <UseAppConfig>false</UseAppConfig>
    <CertStoreLocation>LocalMachine</CertStoreLocation>
    <CertStoreName>Root</CertStoreName>
    <CertFindType>FindBySubjectName</CertFindType>
    <CertFindValue />
  </OperationsCIM2ndEdition>
  <PingVersion>2</PingVersion>
  <OnDemandReadVersion>1</OnDemandReadVersion>
  <UserName>DMS</UserName>
```

```
  <Password>DMS</Password>
  <SystemName>IDMS</SystemName>
  <DemoMeterMode>true</DemoMeterMode>
  <DemoMeterFilePath>AmiServiceMeters.xml</DemoMeterFilePath>
  <MeterPrefix />
  <AutomaticStartup>true</AutomaticStartup>
  <MeterReadInterval>15</MeterReadInterval>
  <EnableDetailedLogging>true</EnableDetailedLogging>
  <FilterMessages>false</FilterMessages>
  <FilterValues>false</FilterValues>
  <VoltHighUnreasonable>NaN</VoltHighUnreasonable>
  <VoltLowUnreasonable>NaN</VoltLowUnreasonable>
  <FlowUnreasonable>NaN</FlowUnreasonable>
  <EventValidTime>15</EventValidTime>
  <EnableNominalVoltageEvent>false</EnableNominalVoltageEvent>
  <VoltageEventTimeoutMinutes>120</VoltageEventTimeoutMinutes>
</AmiConfig>
```

The default `AmiServerConfig.xml` configuration file is shown in AMI Server Configuration File.

3. Landis and Gyr Command Center Interface

This chapter provides details about the Landis and Gyr Command Center Interface events.

3.1. Meter Reading Types

Interface messages use a ReadingType identifier to specify the time, unit, multiplier, and other parameters of the meter readings. The full specification can be found in IEC 61968-9.

3.1.1. Meter Ping

Type	CIM 1	CIM 2
Ping		0.0.0.0.0.1.11.0.0.0.0.0.0.0.0.0.109.0

3.1.2. Meter Readings

Type	CIM 1	CIM 2
Volta Phase A	0.0.12.0.0.0.0.1.129.0.29	
Volta Phase B	0.0.12.0.0.0.0.1.65.0.29	
Volta Phase C	0.0.12.0.0.0.0.1.33.0.29	
Instantaneous kW	0.1.1.1.0.12.0.0.0.3.38	
Instantaneous kVar	0.1.1.1.0.12.0.0.0.3.63	

3.2. Meter Event Types

Interface messages use an EventType identifier to specify the type, phase, and other parameters of the event. The full specification can be found in IEC 61968-9.

3.2.1. Power On/Off Event Type

Type	CIM 1	CIM 2
Power Off	3.26.0.85	
Power On	3.26.0.216	

3.2.2. Sag Started Event Types

Type	CIM 1	CIM 2
Sag Started Phase A	3.38.1.223	
Sag Started Phase B	3.38.5.223	
Sag Started Phase C	3.38.14.223	

3.2.3. Sag Stopped Event Types

Type	CIM 1	CIM 2
Sag Stopped Phase A	3.38.5.224	
Sag Stopped Phase B	3.38.9.224	
Sag Stopped Phase C	3.38.14.224	

3.2.4. Swell Started Event Types

Type	CIM 1	CIM 2
Swell Started Phase A	3.38.1.248	
Swell Started Phase B	3.38.5.248	
Swell Started Phase C	3.38.14.248	

3.2.5. Swell Stopped Event Types

Type	CIM 1	CIM 2
Swell Stopped Phase A	3.38.5.249	
Swell Stopped Phase B	3.38.9.249	
Swell Stopped Phase C	3.38.14.249	

3.2.6. Meter Ping Request

3.2.6.1. Characteristics

Message Type	SOAP/XML
Source	ADMS
Response	Reply and Response

Reading Type	CIM1: CIM2: 0.0.0.0.0.1.11.0.0.0.0.0.0.0.0.0.109.0

3.2.6.2. Sample Request Message


```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns="http://www.landisgyr.com/iec61968/2010/03"
xmlns:mes="http://iec.ch/TC57/2011/schema/message">
  <soapenv:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <RequestMessage xmlns="http://iec.ch/TC57/2011/schema/message">
      <Header>
        <Verb>get</Verb>
        <Noun>MeterReadings</Noun>
        <Revision>2.0</Revision>
        <Timestamp>2014-05-23T09:14:08.9243853-05:00</Timestamp>
        <Source>MDMS</Source>
        <ReplyAddress>http://localhost:8800/mockCIM2ndEdition</ReplyAddress>
        <MessageID>Ping</MessageID>
        <CorrelationID>Ping</CorrelationID>
      </Header>
      <Request>
        <GetMeterReadings xmlns="http://iec.ch/TC57/2011/GetMeterReadings#">
          <EndDevice>
            <Names>
              <name>501D17C9</name>
            </Names>
          </EndDevice>
          <ReadingType>
            <Names>
              <name>0.0.0.0.0.1.11.0.0.0.0.0.0.0.0.0.109.0</name>
            </Names>
          </ReadingType>
        </GetMeterReadings>
      </Request>
    </RequestMessage>
  </soapenv:Body>
</soapenv:Envelope>
```


3.2.6.3. Sample Response Message


```
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <ResponseMessage xmlns="http://iec.ch/TC57/2011/schema/message">
      <Header>
        <Verb>reply</Verb>
        <Noun>MeterReadings</Noun>
        <Revision>2.0</Revision>
```

```
      <Timestamp>2017-03-21T19:01:13.6992369-04:00</Timestamp>
      <Source>command center</Source>
      <MessageID>5eb8ba1b-fa63-4ea9-a77b-ff163571d2f1</MessageID>
      <CorrelationID>fdb1c740-0db4-4ac7-a79d-65848ea9a432</CorrelationID>
    </Header>
    <Reply>
      <Result>OK</Result>
      <Error>
        <code>0.0</code>
      </Error>
    </Reply>
    <Payload>
      <MeterReadings xmlns="http://iec.ch/TC57/2011/MeterReadings#">
        <MeterReading>
          <Meter>
            <Names>
              <name>421857KD</name>
                <NameType>
                  <name>Meter Number</name>
                  <NameTypeAuthority>
                    <name>PPL - DP</name>
                  </NameTypeAuthority>
                </NameType>
            </Names>
          </Meter>
          <Readings>
            <reason>inquiry</reason>
            <reportedDateTime>2017-03-21T19:01:08-04:00</reportedDateTime>
            <source>command center</source>
            <timeStamp>2017-03-21T19:01:08-04:00</timeStamp>
            <value>1</value>
            <ReadingQualities>
              <source>command center</source>
              <timeStamp>2017-03-21T19:01:08-04:00</timeStamp>
              <ReadingQualityType ref="2.0.0"/>
            </ReadingQualities>
            <ReadingType ref="0.0.0.0.0.1.11.0.0.0.0.0.0.0.0.0.109.0"/>
          </Readings>
        </MeterReading>
      </MeterReadings>
    </Payload>
  </ResponseMessage>
 </s:Body>
</s:Envelope>
```

4. Verifying the Integration

This chapter provides procedures for verifying the correct integration and operation of the Landis and Gyr interface.

4.1. Setting Up the Test Environment

The test procedures throughout this chapter assume that the configuration parameters are set to the values as described in Configuration.

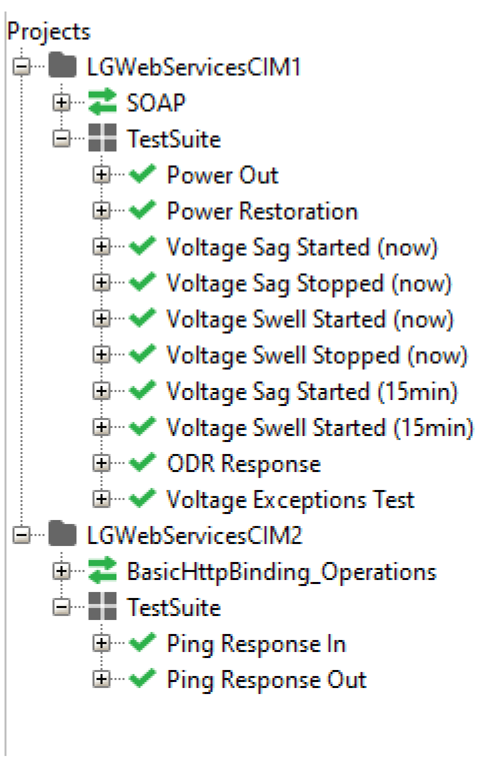
Opening a web service requires either administrator privileges or a URL reservation within Windows. To make URL reservations, run the following commands:

```
netsh http add urlacl url=http://+:8800/CIM2ndEdition user=DOMAIN\User
netsh http add urlacl url=http://+:8800/CIM1stEdition user=DOMAIN\User
```

The following are prerequisites for testing:

- SoapUI is installed.
- The `LGWebServiceCIM1-soapui-project.xml` and `LGWebServiceCIM2-soapui-project.xml` projects are imported into SoapUI.

  To obtain the test SoapUI projects, contact your GE project team.



4.2. Basic Connectivity

The tests throughout this section verify basic Landis and Gyr interface connectivity.

4.2.1. Listening Services Active

Objective: This procedure verifies that web services hosted by AMI Server are available.

Setup: AMI Server is configured for LGWebServices.

Procedure	Results/Verification
1. Start AMI Server.	Verify the process is started.
2. Using a web browser, navigate to http://localhost:8800/CIM1stEdition.	Verify the WCF service help page opens.
3. Using a web browser, navigate to http://localhost:8800/CIM2ndEdition.	Verify the WCF service help page opens.

4.2.2. Endpoint Message Test

Objective: This procedure verifies that web services hosted by AMI Server are available.

Setup:

- AMI Server is configured for LGWebServices.
- The `LGWebServiceCIM1-soapui-project.xml` and `LGWebServiceCIM2-soapui-project.xml` projects are imported into SoapUI.

Procedure	Results/Verification
1. Start AMI Server.	Verify the process is started.
2. Using SoapUI, expand TestSuite in LGWebServicesCIM1.	
3. Double-click Power Out, and then click the URL button on the Power Out window.	Verify the address shown matches the service address (http://localhost:8800/CIM1stEdition).
4. Click the Play button to send the test message.	Verify the AMI Server log shows that it received the message.
5. Using SoapUI, expand TestSuite in LGWebServicesCIM2.	
6. Double-click Ping Response In, and then click the URL button on the Power Out window.	Verify the address shown matches the service address (http://localhost:8800/CIM2ndEdition).
7. Click the Play button to send the test message.	Verify the AMI Server log shows that it received the message.

4.3. Functionality

The tests in this section verify the Landis and Gyr interface functionality.

4.3.1. Meter Ping

Objective: This procedure verifies the ability to ping a single customer from the client using the Customer right-click menu.

Setup: The ADMS geographic network view is open.

Procedure	Results
1. Select a customer in the geographic network view and select the Call Ping Pop-Up - Single Customer option.	Verify that the "CIS/AMI Customer Ping" pop-up display appears and correctly shows data for the selected customer.
2. From the"CIS/AMI Customer Ping" pop-up display, select the Ping checkbox.	Verify that the "Ping", "Deselect All", and "Cancel" buttons are enabled, the "Select All" button is disabled, and the Ping checkbox is selected. The "Ping Status" field shows the current state of the customer.
3. Click "Ping".	Verify that the "Ping Status" field changes to "Sent". Verify that a blue circle appears around the customer symbol.
4. Using SoapUI, expand TestSuite in LGWebServicesCIM2.	
5. Double-click "Ping Response In". Start the test. A "Ping Response" is sent to the AMI Server.	Verify that the ping status changes to "Successful". Verify that the circle around the customer symbol changes from blue to green.

4.3.2. Show Customer Analogs

Objective: This procedure verifies that the single meter data popup display opens correctly via the rick-click menu.

Setup: The ADMS geographic network view is open.

Procedure	Results
1. In SoapUI, expand TestSuite in LGWebServicesCIM1.	
2. Double-click "ODR Response". Start the test. The "Meter Readings Response" is sent to AMI Server.	Verify that the message is sent successfully.
3. Select the previously pinged customer in the geographic network view and select the Show Analogs - Single Customer option.	
4. Left-click the SHOW ANALOGS - SINGLE CUSTOMER" option.	Verify that the "Customer Analogs - Single Customer" pop-up display appears and correctly shows the selected customer’s data.

4.3.3. AMI State Visualization - Voltage Exceptions

Objective: This procedure verifies the visualization of the Low, High, and Return to Normal Voltage Exception functionality.

Setup:

- The ADMS geographic network view is open.
- In `AMIServerConfig.xml`, set the `EventValidTime` parameter to 15.
- In `NetServerMainConfig.txt` and `NetServerOmsConfig.txt`, set the `AMI_STATUSES_TIMEOUT` parameter to 60.

Procedure	Results/Verification
1. From the geographic network view, navigate to a customer and note the customer’s meter number.	Verify that the customer is shown without any halo or symbol.
2. Using SoapUI, expand TestSuite in LGWebServicesCIM1.	
3. Double-click "Voltage Sag Started (now)". Start the test. The "Voltage Sag Started" event is sent to AMI Server.	Verify, after a period of time, that a low voltage exception halo (brown house symbol) appears for the customer.
4. Wait for 60 seconds, as specified by the "AMI_STATUSES_TIMEOUT" parameter.	Verify that the low voltage exception halo disappears.
5. Open "Voltage Sag Stopped (now)". Start the test. The "Voltage Sag Stopped" event is sent to AMI Server.	Verify that no meter exception status halo shows for the target customer.
6. Open "Voltage Swell Started (now)". Start the test. The "Voltage Swell Started" event is sent to AMI Server.	Verify, after a period of time, that a high voltage exception halo (yellow house symbol) appears for the customer.
7. Wait for 60 seconds, as specified by the "AMI_STATUSES_TIMEOUT" parameter.	Verify that the low voltage exception halo disappears.
8. Open "Voltage Swell Stopped (now)". Start the test. The "Voltage Swell Stopped" event is sent to AMI Server.	Verify that no meter exception status halo shows on the target meter.
9. Open "Voltage Sag Started (15 min). Start the test. A "Voltage Sag Started" event with a timestamp greater than 15 minutes in the past is sent to AMI Server.	Verify that no meter exception status halo shows on the target meter. Verify in the AMI Server log that the message was received and discarded.
10. Open "Voltage Swell Started (15 min)". Start the test. A "Voltage Swell Started" event with a timestamp greater than 15 minutes in the past is sent to AMI Server.	Verify that no meter exception status halo shows for the target customer. Verify in the AMI Server log that the message was received and discarded.

4.3.4. AMI State Visualization - Power On/Off

Objective: This procedure verifies visualization of Meter Outage/Restore events.

Setup:

- The ADMS geographic network view is open.
- In `NetServerMainConfig.txt` and `NetServerOmsConfig.txt`, set the AMI_STATUSES_TIMEOUT parameter to 60.

Procedure	Results/Verification
1. In SoapUI, expand TestSuite in LGWebServicesCIM1.	

Procedure	Results/Verification
2. Double-click "Power Out". Start the test. A "Power Out" event is sent to AMI Server.	Verify, after a period of time, that a meter outage halo (red square) appears as the target customer’s symbol.
3. Wait for 60 seconds, as specified by the "AMI_STATUSES_TIMEOUT" parameter.	Verify that the meter outage halo disappears.
4. Open "Power Restoration". Start the test. A "Power Restoration" event is sent to AMI Server.	Verify, after a period of time, that a meter restoration halo (green square) appears as the target customer’s symbol.
5. Wait for 60 seconds, as specified by the "AMI_STATUSES_TIMEOUT" parameter.	Verify that the meter restoration halo disappears.
6. From the geographic network view toolbar, turn on the "Customers with Activity" customer layer.	Verify that the target customer and the meter outage/restoration halos are not shown in the display.
7. In SoapUI, double-click "Power Out". Start the test. A "Power Out" event is sent to AMI Server.	Verify, after a period of time, that the meter outage halo appears as the target customer’s symbol. Customers with "Call events" and "Meter Outage/Restore events" are considered as customers with activity.
8. From the geographic network view toolbar, turn on the "Customers with Activity" customer layer.	Verify that the target customer and the meter outage/restoration halos are not shown in the display.
9. In SoapUI, open "Power Restoration". Start the test. A "Power Restoration" event is sent to AMI Server.	Verify, after a period of time, that the meter restoration halo (green square) appears as the target customer’s symbol. Customers with "Call events" and "Meter Outage/Restore events" are considered as customers with activity.

5. AMI Server Configuration File

The `AmiServerConfig.xml` configuration file is stored in `%IDMS_ROOT%\config\server\`. The default configuration is as follows:

```
<?xml version="1.0" encoding="utf-8"?>
<CallServerConfig>
  <ApplicationParameters>
    <CustomerFile>%IDMS_ROOT%\models\customer\FantasyIslandCustomers.bpcl</CustomerFile>
    <!-- Relative location (path) of the customer file-->
    <CheckMessageSecurity>true</CheckMessageSecurity>
    <!--If set to true this parameter will enable checking of security token authenticity/expiration--
>
    <UseCredentialFile>false</UseCredentialFile>
    <!--If set to true this parameter will enable use of the Credential file for defining the private
key and user SIDs-->
    <AllowUnregisteredSids>true</AllowUnregisteredSids>
    <!--If set to true this parameter will enable any user SID to be valid-->
    <Relay>true</Relay>
    <!--Set to true for the Relay Server or the AMI Server.  For Call Server this parameter must be
false.-->
    <CallServerPort>5032</CallServerPort>
    <!--TCP/IP listening port number of the Call Server-->
    <CallServerHost>localhost</CallServerHost>
    <!--Host machine name of the Call Server-->
    <LogPath>.</LogPath>
    <!--Relative path where the log file of the Call Server will be saved-->
    <LogDetails>false</LogDetails>
    <!--Controls whether detail (debug) log messages are written to the log-->
    <ServerLogFileName>%IDMS_ROOT%\log\AMIServer.log</ServerLogFileName>
    <!--Name and path to the CallServer log file.-->
    <ServerLogEnabled>true</ServerLogEnabled>
    <IsServiceMode>true</IsServiceMode>
    <!--Set to true only for the AMI Server.  For Call Server and Relay Server this parameter must be
false.-->
    <CIMInteropEnabled>true</CIMInteropEnabled>
    <!--Enable the CIM Interop web service (Applies to AMI Server only).-->
    <AmiServerJournal>%IDMS_ROOT%\dynamics\server\AmiServer\</AmiServerJournal>
    <ItronPowerOffString>Primary Power Down</ItronPowerOffString>
    <ItronPowerOnString>Primary Power Up</ItronPowerOnString>
    <ItronGroupName>Std_1Ch5</ItronGroupName>
    <DisableItronInterface>false</DisableItronInterface>
    <CredentialInfoFile>%IDMS_ROOT%\config\server\CallServerCredentialInfo.xml</CredentialInfoFile>
    <!--File name and path of the Credential Info file that stores the valid SIDs and secret shared
key-->
    <!--Ami Interface Plugin-->
    <EnableAmiInterface>false</EnableAmiInterface>
    <!-- Interface name (displayed on UI Tab) -->
    <AmiInterfaceName>LGCIM</AmiInterfaceName>
    <!-- Path to plugin dll -->
    <AmiInterfaceDllPath>%DMS_BINDIR%\LGWebServices.dll</AmiInterfaceDllPath>
    <!-- Type which implements CallServerAmiInterface -->
```

```
    <AmiInterfaceDllTypeName>AMIWebServices.LGWebServices</AmiInterfaceDllTypeName>
  </ApplicationParameters>
  <SystemCapabilities>
    <SupportsCaseNoteQuery>true</SupportsCaseNoteQuery>
    <!--true if the system supports case note query-->
    <SupportsMeterEventQuery>false</SupportsMeterEventQuery>
    <!--true if the system supports meter event query-->
    <SupportsMeterPingRequest>true</SupportsMeterPingRequest>
    <!--true if the system supports meter ping request-->
    <SupportsCallQuery>false</SupportsCallQuery>
    <!--true if the system supports call query-->
    <SupportsAutomatedCallbackRequest>false</SupportsAutomatedCallbackRequest>
    <!--true if the system supports automatic callback request-->
    <SupportsManualCallbackRequest>false</SupportsManualCallbackRequest>
    <!--true if the system supports manual callback request-->
  </SystemCapabilities>
  <ItronConfig>
    <!-- Subscription paramters -->
    <SystemName>IDMS</SystemName>
    <MeterPrefix>2.16.840.1.114416.0.</MeterPrefix>
    <DemoMeterMode>true</DemoMeterMode>
    <SubscribeOnStartup>true</SubscribeOnStartup>
    <ExceptionSubscriptionEnabled>true</ExceptionSubscriptionEnabled>
    <ExceptionSubscriptionGroupNames>
      <string>Demo</string>
    </ExceptionSubscriptionGroupNames>
    <!--Unknown, Configuration, StatusEvent, BillingEvent, CommonOperation, PowerOutageOrRestoration,
Communication, HanDeviceEvent-->
    <ExceptionSubscriptionCatergoryContraints>
      <ExceptionCategory>StatusEvent</ExceptionCategory>
      <ExceptionCategory>PowerOutageOrRestoration</ExceptionCategory>
    </ExceptionSubscriptionCatergoryContraints>
    <DataSubscriptionEnabled>true</DataSubscriptionEnabled>
    <DataSubscriptionGroupNames>
      <string>Demo</string>
    </DataSubscriptionGroupNames>

    <!-- Message strings -->
    <Whd>Wh d</Whd>
    <Whr>Wh r</Whr>
    <VARhd>VARh d</VARhd>
    <VARhr>VARh r</VARhr>
    <Vha>Vh (a)</Vha>
    <Vhb>Vh (b)</Vhb>
    <Vhc>Vh (c)</Vhc>
    <PrimaryPowerDown>Primary Power Down</PrimaryPowerDown>
    <PrimaryPowerUp>Primary Power Up</PrimaryPowerUp>
    <VoltHourAboveThreshold>Vh Above Threshold</VoltHourAboveThreshold>
    <VoltHourBelowThreshold>Vh Below Threshold</VoltHourBelowThreshold>
    <VoltHourReturnToNormal>Vh Restored From Below Threshold to Nominal</VoltHourReturnToNormal>
    <MeterReadInterval>5</MeterReadInterval>
    <EnableDetailedLogging>true</EnableDetailedLogging>
    <!-- Data Conversion -->
```

```
    <PolyPhaseMeters>
        <string>73.84.82.701</string>
        <string>73.84.82.52 </string>
    </PolyPhaseMeters>
    <DataConversionDll>ItronAmiMessageProcessor.dll</DataConversionDll>
    <DataConversionType>ItronAmiMessageProcessor.ItronAmiMessageProcessor</DataConversionType>
    <VoltHighUnreasonable>999.99</VoltHighUnreasonable>
    <VoltLowUnreasonable>0</VoltLowUnreasonable>
    <FlowUnreasonable>999.99</FlowUnreasonable>
  </ItronConfig>
  <AmiConfig xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance">
    <!-- Listening service addresses -->
    <EnableEventListenerServiceCIM1stEdition>true</EnableEventListenerServiceCIM1stEdition>

<EventListenerServiceAddress1stEdition>https://localhost/CIM1stEdition</EventListenerServiceAddress1st
Edition>
    <EnableEventListenerServiceCIM2ndEdition>true</EnableEventListenerServiceCIM2ndEdition>

<EventListenerServiceAddress2ndEdition>https://localhost/CIM2ndEdition</EventListenerServiceAddress2nd
Edition>
    <!-- Demo meter mode maps external meter numbers to intermal meter numbers -->
    <DemoMeterMode>false</DemoMeterMode>
    <!-- Path to the meter mapping file -->
    <DemoMeterFilePath>AmiServiceMeters.xml</DemoMeterFilePath>
    <!-- System name to include in messages -->
    <SystemName>IDMS</SystemName>
    <!-- MeterPrefix to be added to all external communication -->
    <MeterPrefix/>
    <!-- Connect on startup-->
    <AutomaticStartup>true</AutomaticStartup>
    <!-- Log all messages -->
    <EnableDetailedLogging>true</EnableDetailedLogging>
    <!-- Operations CIM1 Client settings -->
    <OperationsCIM1stEdition>
      <BaseAddress>https://www.lgwebservice.com/GSIS_APIService/cim_2_0/CIMService.svc</BaseAddress>
      <UserName>dms</UserName>
      <Password>DMS</Password>
      <PingReadingType>0.0.0.0.0.1.11.0.0.0.0.0.0.0.0.0.109.0</PingReadingType>
      <OnDemandReadingTypes/>
      <!-- Reply address should point to EventListenerServiceAddress1stEdition -->
      <ReplyAddress>https://localhost/CIM1stEdition</ReplyAddress>
      <!-- true to use CallServer.exe.config -->
      <UseAppConfig>false</UseAppConfig>
      <CertStoreLocation>CurrentUser</CertStoreLocation>
      <CertStoreName>Root</CertStoreName>
      <CertFindType>FindBySubjectName</CertFindType>
      <CertFindValue/>
    </OperationsCIM1stEdition>
    <!-- Operations CIM2 Client settings -->
    <OperationsCIM2ndEdition>
      <BaseAddress>https://www.lgwebservice.com/GSIS_CIM2ndEdition/CIM2ndEd.svc</BaseAddress>
      <UserName>dms</UserName>
```

```
      <Password>DMS</Password>
      <PingReadingType>0.0.0.0.0.1.11.0.0.0.0.0.0.0.0.0.109.0</PingReadingType>
      <OnDemandReadingTypes>
        <string>0.0.0.1.19.1.12.0.0.0.0.0.0.0.0.3.72.0</string>
        <string>0.0.0.1.1.1.12.0.0.0.0.0.0.0.0.3.72.0</string>
      </OnDemandReadingTypes>
      <!-- Reply address should point to EventListenerServiceAddress2ndEdition -->
      <ReplyAddress>https://WIN-DMSINT-P2.ppl.com/CIM2ndEdition</ReplyAddress>
      <UseAppConfig>false</UseAppConfig>
      <CertStoreLocation>CurrentUser</CertStoreLocation>
      <CertStoreName>Root</CertStoreName>
      <CertFindType>FindBySubjectName</CertFindType>
      <CertFindValue/>
    </OperationsCIM2ndEdition>
    <UserName />
    <Password />
    <!-- Enable return to normal, or nominal voltage events -->
    <EnableNominalVoltageEvent>false</EnableNominalVoltageEvent>
    <!-- Automatic voltage nominal events are created after no events are received for a meter.  0 =
disabled -->
    <VoltageEventTimeoutMinutes>120</VoltageEventTimeoutMinutes>
    <!-- Maximum valid reading value for Volts -->
    <VoltHighUnreasonable>288</VoltHighUnreasonable>
    <!-- Minimum valid reading value for Volts -->
    <VoltLowUnreasonable>20</VoltLowUnreasonable>
    <!-- Maximum valid reading value for Watt and Var -->
    <FlowUnreasonable>500</FlowUnreasonable>
    <!-- Discard the entire message is any value is unreasonable -->
    <FilterMessages>false</FilterMessages>
    <!-- Discard only those values which are unreasonable -->
    <FilterValues>true</FilterValues>
    <!-- Maximum number of minutes before now that incoming events will still be processed -->
    <EventValidTime>15</EventValidTime>
    <!-- Customer File column to use for ReadingMultiplier calculations -->
    <ReadingMultiplierColumnName>PTRatio</ReadingMultiplierColumnName>
    <!-- Default meter number to use on the LGCIM tab for testing -->
    <TestMeterNumber>300089419</TestMeterNumber>
    <!-- Calculate Var based on a set PowerFactor -->
    <CalculateVar>true</CalculateVar>
    <!-- Power Factor value to use in calculations -->
    <PowerFactor>.85</PowerFactor>
    <!-- Create event groups within AMI Server by caching events -->
    <EventCacheEnabled>true</EventCacheEnabled>
    <!-- Seconds to wait before processing the "grouped" messages -->
    <EventCacheTime>30</EventCacheTime>
    <!-- Enable the MeterReading Mappings -->
    <ProcessMeterReadingWithIndex>true</ProcessMeterReadingWithIndex>
    <!-- Map a MeterReading to a different ReadingType than in the message, based on the
MeterReading's order in the message -->
    <MeterReadingMapping>
      <MeterReadingIndexMapping>
        <Index>6</Index>
        <ReadingType>0.1.1.1.0.12.0.0.0.0.38</ReadingType>
```

```
      </MeterReadingIndexMapping>
      <MeterReadingIndexMapping>
        <Index>7</Index>
        <ReadingType>0.1.1.1.0.12.0.0.0.0.63</ReadingType>
      </MeterReadingIndexMapping>
    </MeterReadingMapping>
    <!--
      EventTypes for reference
    -->
    <PowerOffEventType>3.26.0.85</PowerOffEventType>
    <PowerOnEventType>3.26.0.216</PowerOnEventType>
    <SagStartedPhaseAEventType>3.38.1.223</SagStartedPhaseAEventType>
    <SagStartedPhaseBEventType>3.38.5.223</SagStartedPhaseBEventType>
    <SagStartedPhaseCEventType>3.38.14.223</SagStartedPhaseCEventType>
    <SagStoppedPhaseAEventType>3.38.5.224</SagStoppedPhaseAEventType>
    <SagStoppedPhaseBEventType>3.38.9.224</SagStoppedPhaseBEventType>
    <SagStoppedPhaseCEventType>3.38.14.224</SagStoppedPhaseCEventType>
    <SwellStartedPhaseAEventType>3.38.1.248</SwellStartedPhaseAEventType>
    <SwellStartedPhaseBEventType>3.38.5.248</SwellStartedPhaseBEventType>
    <SwellStartedPhaseCEventType>3.38.14.248</SwellStartedPhaseCEventType>
    <SwellStoppedPhaseAEventType>3.38.5.249</SwellStoppedPhaseAEventType>
    <SwellStoppedPhaseBEventType>3.38.9.249</SwellStoppedPhaseBEventType>
    <SwellStoppedPhaseCEventType>3.38.14.249</SwellStoppedPhaseCEventType>
  </AmiConfig>
</CallServerConfig>
```